

CODI vs KaVa: A Controlled Study of Latent Reasoning Supervision

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Abstract. This study asks whether KaVa's key/value (KV) trajectory supervision improves continuous latent mathematical reasoning over CODI's endpoint hidden-state distillation under matched conditions. Both methods were implemented in one GPT-2 framework and controlled for data, architecture, six autoregressive latent steps, optimizer, schedule, decoding, and a one-epoch budget of 96,405 steps. On the complete seed-zero evaluation, KaVa increased macro numeric exact-match accuracy from **11.28%** to **13.29%**: a gain of **2.01 percentage points** (95% paired-bootstrap CI: +0.44 to +3.60). KaVa also outperformed CODI at all three matched seeds. Cross-example latent-state shuffling harmed KaVa 4.60 points more than CODI, showing stronger dependence on example-specific latent information. The advantage is reliable but task-dependent and concentrated on MultiArith; current controls do not isolate R-KV compression as its unique cause.

1. Research questions and controlled design

The broad question, *Which method is better?*, was refined into three tests: **(1)** Does KaVa outperform CODI under matched conditions? **(2)** Is the direction consistent across training seeds? **(3)** Does KaVa make stronger causal use of its latent states? CODI and KaVa share student and teacher cross-entropy plus endpoint hidden-state matching. KaVa additionally matches the student's six latent KV states to a teacher KV trajectory compressed with R-KV.

Controlled component	Setting	Evaluation	Setting
Backbone / data	GPT-2 / 385,620 examples	Metric	Numeric exact match
Latent budget	6 autoregressive states	Full sample	3,118 items per method
Optimization	Batch 4; LR 1e-4	Uncertainty	10,000 paired bootstraps
Training budget	1 epoch; 96,405 steps	Decoding	Greedy; max 64 tokens

2. Complete primary evaluation

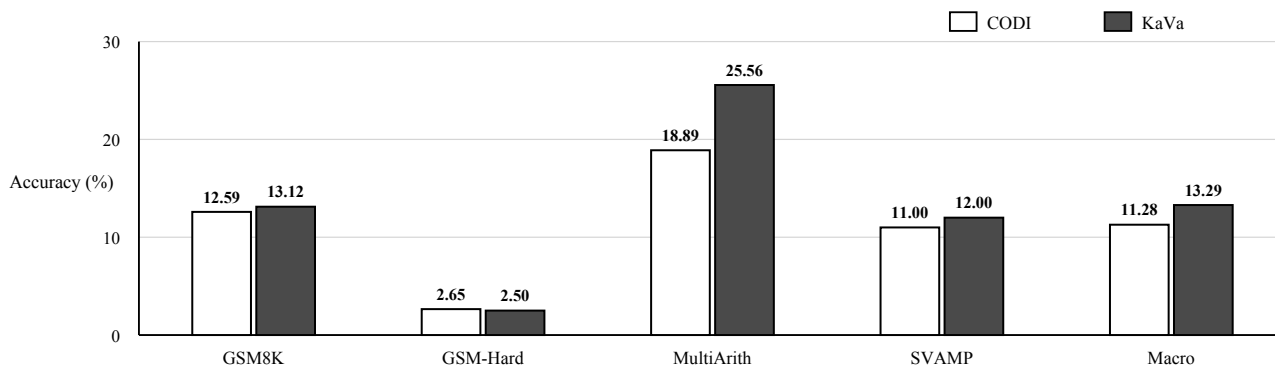


Figure 1. Complete seed-zero numeric exact-match accuracy. Bars show percentages; macro is the unweighted mean of the four tasks. KaVa improves macro accuracy by **+2.01 points** (95% CI +0.44 to +3.60). MultiArith supplies most of the gain: **+6.67 points** (95% CI +1.11 to +12.22; McNemar $p = 0.02896$).

Primary result	KaVa is 17.8% higher relative to CODI's macro accuracy, and the complete paired interval excludes zero.
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3. Matched-seed replication

On capped matched evaluations, KaVa outperformed CODI at seeds 0, 1, and 2 by +2.17, +0.97, and +1.08 macro points. The mean advantage was +1.41 points (sample SD 0.66). This establishes directional consistency across the tested initializations, while three seeds remain insufficient for a precise population-level method interval.

4. Compression controls

Seed-zero capped run	Macro	Scientific role
CODI	11.72%	Endpoint hidden-state distillation
No trajectory distillation	13.19%	Tests whether trajectory matching is necessary
KaVa - random compression	13.40%	Removes learned R-KV selection
KaVa - uniform compression	12.69%	Uses fixed temporal coverage
KaVa - R-KV	13.89%	Full configuration; numerically best

Full KaVa is numerically best, but its paired macro differences from the three controls have intervals that include zero. The evidence therefore supports the complete KaVa configuration over CODI, but does not attribute the improvement uniquely to R-KV compression.

5. Causal evidence

Causal interventions replaced latent states in both models. The difference-in-differences statistic is (KaVa intervention - KaVa baseline) - (CODI intervention - CODI baseline); negative values mean that the same corruption harmed KaVa more.

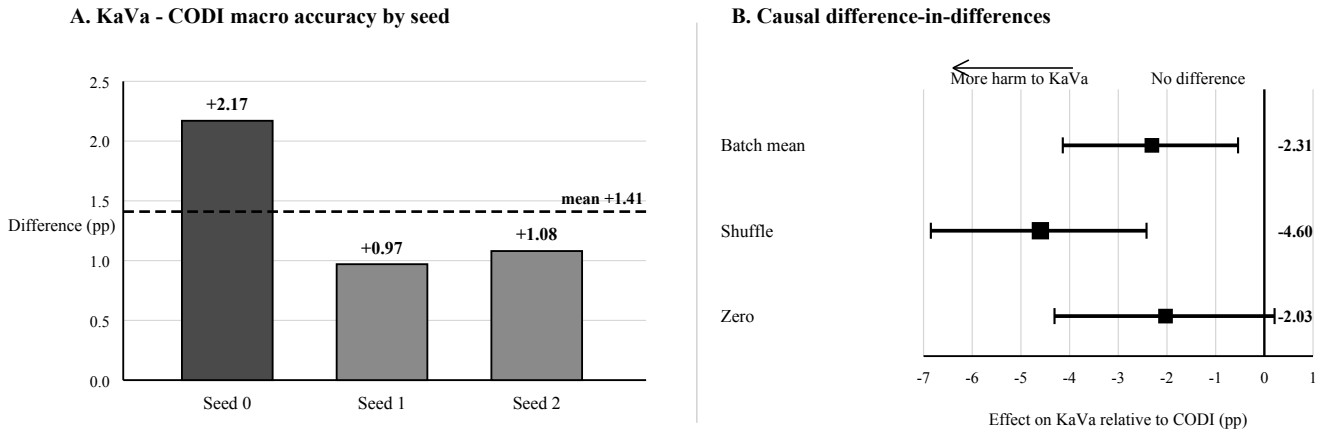


Figure 2. A, KaVa's matched-seed macro advantage was +2.17, +0.97, and +1.08 points (mean +1.41; sample SD 0.66). B, causal effects with 95% paired-bootstrap intervals. Cross-example shuffling produced the largest differential harm: -4.60 points (95% CI -6.85 to -2.42), demonstrating stronger use of example-specific latent information by KaVa.

Position-wise shuffling localized KaVa's strongest dependency to latent position 4, where capped macro accuracy fell from 13.89% to 10.26%.

6. Conclusion and research direction

All three primary questions receive affirmative answers in this controlled setting: KaVa is more accurate overall; it improves over CODI at each of three matched seeds; and it relies more strongly on example-specific latent states. The gain is modest and task-dependent, with MultiArith as the main driver. Explicit CoT-SFT remains a stronger absolute reference at 33.86% macro accuracy.

The next high-value question is whether latent states implement causally structured reasoning or mainly supply additional serial computation. A compute-matched study should compare endpoint hidden states, keys-only, values-only, full KV trajectories, and unsupervised recurrent computation, followed by targeted activation patching. In parallel, CODI versus KaVa can anchor a systematic survey organized by representation, supervision granularity, transition mechanism, causal evidence, and accuracy-efficiency trade-offs.

Limitations

The study uses GPT-2, arithmetic tasks, six latent positions, and one training epoch. Only the primary seed-zero pair has complete evaluation; other runs use capped sets. Three seeds establish directional consistency but not a precise population-level confidence interval.

References

- [1] Z. Shen et al., "CODI: Compressing Chain-of-Thought into Continuous Space via Self-Distillation," arXiv:2502.21074, 2025.
- [2] A. Kuzina et al., "KaVa: Latent Reasoning via Compressed KV-Cache Distillation," arXiv:2510.02312, 2025.